

Matthias Fitzky, PhD

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Open to relocation | Passport: Germany

Geospatial Data Scientist | Earth Observation & GeoAI | GIS, Remote Sensing, Climate Risk Modeling

Skills:

Data Analytics & Machine Learning: Python (NumPy, Pandas, scikit-learn, PyTorch), SQL, Oracle DB, PostgreSQL; regression and clustering, feature engineering, dimensionality reduction (PCA, autoencoders), time-series analysis, model validation, Bayesian reasoning and uncertainty quantification, data visualization (Matplotlib, Seaborn), ETL workflows (Airflow)

GeoAI & Remote Sensing: QGIS, ArcGIS Pro, Google Earth Engine, GeoPandas, FME; raster and vector data (GeoTIFF, Shapefile, GeoJSON), satellite imagery (Landsat, Sentinel, NAIP), LiDAR; photogrammetry (camera calibration, image orientation, point-cloud processing, orthorectification workflows using Pix4D, Agisoft Metashape, Trimble Inpho Suite); 3D city models, spatial statistics, PostGIS

Programming & Data Engineering: Git, Linux/Bash, workflow management (SLURM), Docker, GDAL/PROJ, data structuring and cleaning

Modeling & Visualization: ANSYS Fluent (CFD), Tecplot, ParaView, AutoCAD (Civil 3D); dashboard design, technical reporting

IoT & Sensor Data: Raspberry Pi sensor systems, IoT data acquisition, environmental monitoring, data quality control

GitHub Portfolio: selected projects in geospatial analytics, predictive modeling, and physics-informed ML

Education:

New York University

New York City, USA

PhD in Urban Systems

Sep. 2022 – Jan. 2026 (completed)

Dissertation Title: A Computational Framework for Rapid Prediction of Human Thermal Exposure in Complex Built Environments

Focus: Urban microclimate modeling, geospatial analysis, integration of environmental sensor and simulation data, physics-informed machine learning

University of Applied Sciences Stuttgart

Stuttgart, Germany

Master of Science in Photogrammetry and Geoinformatics

Feb. 2018 – Aug 2019

Master's Thesis: Simulation of Cooling Energy Demand Using the 3D City Model of Singapore

Focus: 3D geospatial data, Photogrammetry & Remote Sensing, urban energy modeling

Bachelor of Engineering in Land Surveying and Geoinformatics

Sep. 2014 – Jan. 2018

Bachelor's Thesis: Comparative Analysis of Different Methods and Datasets for Calculating Building Energy Certificates

Experience:

New York University

Research Fellow & PhD Researcher

Abu Dhabi, UAE & New York City, USA

Jan. 2020 – Jan. 2026

Geospatial Analysis & Remote Sensing

- Assessed disaster/conflict impacts using VIIRS nightlight and radar/LiDAR (radiance changes, land subsidence); modeled flood and SLR scenarios with Copernicus DSM and USGS LiDAR.
- Analyzed urban heat drivers via ElasticNet regression; mapped NDVI trends in GEE to quantify vegetation change across NYC (2007-2015).
- Evaluated rooftop solar potential; built 3D photogrammetric city models from UAV/aerial imagery and LiDAR point clouds for urban morphology and facade analysis.

Machine Learning on Spatial & Environmental Data:

- Developed multimodal CNNs predicting urban air temperature from Street View and satellite imagery; built physics-informed ML surrogates linking meteorological inputs to CFD-derived microclimate fields.
- Implemented U-Net and Random Forest workflows for building footprint mapping and urban function classification (NAIP, PLUTO); built a mold-risk index validated against public health data.

Urban Microclimate Modeling:

- Ran high-resolution CFD multiphysics simulations in ANSYS Fluent (NYC, Abu Dhabi, Barcelona) resolving airflow, radiation, and thermal comfort at sub-meter scale; validated against weather stations and thermal imagery.
- Developed ML surrogates (PCA, Random Forests, Autoencoders) for 2D microclimate prediction, cutting simulation time ~200x

Teaching & Mentoring:

- Lab instructor for graduate course Monitoring Cities (27 students): GIS, remote sensing, LiDAR, photogrammetry, and spatial analytics using QGIS, GEE, Pix4D, and LidR.

Singapore Land Authority

Master Thesis Internship

Singapore

Mar. 2019 – Aug. 2019

- Developed a GIS-based workflow to estimate cooling energy demand in Singapore using detailed 3D geospatial datasets and climate-adapted simulation standards.
- Cleaned and enriched 3D geospatial data for integration into urban energy models, aligning building characteristics with local construction and usage types.
- Conducted sensitivity analyses to examine how data quality and modeling assumptions affect urban energy assessments.
- Collaborated with urban planning and smart city researchers at local academic and research institutions to assess geospatial tools for optimizing energy-efficiency in urban building stock.

University of Applied Sciences Stuttgart

Research Assistant

Stuttgart, Germany

Feb. 2017 – Nov. 2019

- Contributed to EU-funded research on urban-scale energy modeling using 3D GIS datasets in partnership with the Joint Research Centre (JRC) of the European Commission.
- Supported data processing and coordination of cross-country case studies across Germany, Italy and Switzerland.
- Co-authored peer-reviewed publications on comparative approaches of energy performance labeling and simulation accuracy

Fichtner IT Consulting GmbH

Working Student

Stuttgart, Germany

Mar. 2017 – Nov. 2017

- Supported the migration and modernization of GIS-based utility infrastructure databases for engineering and municipal clients.
- Used SQL to clean, transform and restructure spatial datasets across different database systems.

Languages:

English (Full Professional Fluency), German (Native), French (Basic Proficiency)

Honors & Awards:

- Runner-Up, Poster Competition, Department of Civil & Urban Engineering, NYU Tandon School of Engineering (2025) – *Modeling Human Thermal Experience in the Urban Microclimate*
- Best Research Presentation, KAIST–NYU Urban-X Young Researcher Day (2023) – “*Data-Driven Simulation of the Urban Microclimate*”
- 1st Place, Tandon Research Excellence Exhibit (2023) – Best project in the VIP category